

INTERNET WITH MOBILE IP

Mobile IP—an extension to the Internet protocol (IP)—supports Internet mobility. How it works and how will it change with IP version 6? Let's see



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The Internet with mobile Internet protocol (IP) is coming in a big way. Two trends augmenting it are the increase in the number of Internet users (Fig. 1) and popularity of laptops and palmtops. It is the expanding scope and application of the Internet that has resulted in the growth of the Internet user base and access.

Of all the Internet users, telecommuters, mobile managers and other remote users find mobile computers appealing, as mobile IP helps them stay seamlessly connected, independent of location. This is unlike to getting the Internet connection on a plugged attachment from location to location. Although mobile IP is particularly suited to wireless connection, it supports both the wireless and wired media.

Basically, the Internet functions with a four-layered architecture (Fig.

2). TCP (Transmission Control Protocol) or IP is the main protocol for the Internet. Mobile IP is a protocol proposed by the Internet Engineering Task Force (IETF) to support mobility in Internet connectivity.

In fact, mobile IP is the next evolution of mobile voice communication. Advances in wireless network technology have revolutionised the method, perception, quality and characteristics of voice communications. Consequently, the developments in wireless data protocols have paved the way for mobile IP for data communications.

What the cellular technology GSM (Global System for Mobile communication) did for telephony, mobile IP will do for the TCP-/IP-based mode of data transport that has become ubiquitous in wireline environments. This type of mobility is already available today in some forms using satellite

access to an Internet service provider (ISP). But satellites provide access to just one ISP at a time—and only as long as the user is within the satellite's footprint. Mobile IP provides a way for carriers to offer assured, continuous access to the Internet or a private IP-based network.

Why the need arises?

IP address is the tool of Internet connectivity. Internet routers pass any received IP datagram to the destination. (IP datagram is a packet with data plus headers fields of different layers of the Internet architecture. This is like a letter that contains the message plus other data like the sender's and receiver's names, addresses and pin codes.) IP datagram carries both the source and destination addresses of data. Thus if the source or destination changes, the connection fails until and unless the changed IP address is notified to all users.

In conventional wired Internet connectivity, when a user connects to the Internet, the point of attachment is made through an address known as IP address. This IP address is fixed—one that's been permanently assigned to his computer.

In wireless connectivity, the users are connected with an IP address that is valid over the coverage area cell of the wireless connectivity.

As a mobile user changes his location dynamically with time, the plugged-in fixed IP address of the wired Internet will be of no use to support connectivity to him. On the other hand, there is no guarantee that the mobility of the user will be limited to the coverage area. So the IP address has to change when the mobile user